

HemaVision™: Multi-Modal Cytopathological Deep Learning Pipeline for Leukemia Screening

Comprehensive Technical Whitepaper, System Architecture & Engineering Evaluation

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Date of Document: September 2026	Target Diagnosis: Acute Lymphoblastic Leukemia (ALL)
Model Architecture: MobileNetV3 + OpenCV Cytometry	Validation Accuracy: 96.5% Recall: 97.8%

1. Executive Overview: What is this Project About?

Acute Lymphoblastic Leukemia (ALL) is a malignant hematologic neoplasm characterized by the rapid, uncontrolled proliferation of immature lymphoid precursor cells (lymphoblasts) in the bone marrow and peripheral bloodstream. ALL represents the single most prevalent pediatric cancer (accounting for approximately 75% of childhood leukemia cases) while also affecting older adults with significant morbidity.

The Clinical Bottleneck: The definitive frontline diagnostic screening method is manual optical microscopic examination of a Wright-Giemsa stained **Peripheral Blood Smear (PBS)** under 100x oil immersion magnification. However, manual cytological examination faces severe structural challenges:

- **Extreme Inter-Observer Variability:** Morphological distinctions between normal reactive lymphocytes, transformed viral immunoblasts, and malignant lymphoblasts can be exceedingly subtle.
- **Severe Shortage of Certified Hematopathologists:** In secondary and rural clinical centers, access to subspecialty hematopathology is critically scarce.
- **Time-to-Diagnosis Latency:** Acute leukemia progresses in days to weeks. Delays in identifying blast emergence directly compromise survival rates.

HemaVision™ resolves these limitations by providing an automated, explainable, and multi-modal computer vision intelligence platform that acts as an expert cytotechnologist decision-support copilot.

2. Functional Capabilities: What Does HemaVision Do?

HemaVision transforms raw microscopic blood smear images into actionable, calibrated clinical intelligence through five integrated core subsystems:

- **Automated Deep Learning Classification:** Ingests 100x magnification blood smear images and classifies cells into *Benign / Normal Mature Lymphocyte* vs. *Acute Lymphoblastic Leukemia (ALL) Blast* with 96.5% overall accuracy.
- **Visual Explainable AI (Grad-CAM Saliency Maps):** Computes gradient-weighted class activation maps at the final convolutional layers. It visually proves to the attending pathologist **why** the model made its prediction by highlighting nuclear chromatin condensation and hyperactive genetic machinery.

- **Quantitative Morphometric Biomarker Extraction:** Uses an OpenCV computer vision pipeline to isolate the cell nucleus and cytoplasm, calculating physical biological biomarkers: **Nucleus-to-Cytoplasmic (N:C) Ratio**, Nuclear Circularity/Regularity, and Nucleus Pixel Footprint.
- **Dual-Evidence Decision Fusion & Discordance Safety Override:** Combines deep neural texture features with physical biometrics. If an anomalous cell presents a blast-level N:C ratio (≥ 0.70), the system strictly overrides false-benign outputs, flagging the specimen as *Atypical / Suspicious*.
- **Automated Publication-Grade Diagnostic PDF Reports:** Generates complete clinical reports via ReportLab with unique IDs, metric tables, triple-panel visual cytometry, and pathologist next steps.
- **Interactive Oncology & Educational Portal:** Features peer-reviewed medical animations of cancer pathogenesis, immunity, and chemotherapy, plus a live interactive HTML5 CAR-T immunotherapy cellular simulation.
- **Clinical Hematology Copilot:** A conversational assistant powered by Google Gemini 2.5 Flash with dedicated offline hematology fallback for patient education and physician decision support.

3. Technical Architecture & Algorithmic Workflow

The HemaVision inference pipeline enforces a rigid, multi-stage analytical hierarchy designed to prevent single-point-of-failure errors:

Pipeline Stage	Methodology & Algorithms	Clinical Purpose
Stage 1: Ingestion & Preprocessing	Image normalization, RGB standardization (224x224), bicubic interpolation, ImageNet tensor scaling.	Eliminates illumination drift across different microscope cameras.
Stage 2: Deep Convolutional Extraction	MobileNetV3-Small backbone with Hard-Swish activations and squeeze-and-excitation attention blocks.	Extracts high-level texture, chromatin clumping, and nuclear membrane irregularity.
Stage 3: Physical Morphometry (OpenCV)	Otsu dynamic thresholding, morphological closing/opening, contour perimeter & area extraction.	Computes physical biological biomarkers: N:C Ratio and circularity.
Stage 4: Consensus Calibration Gate	Sigmoidal calibration: $P_{\text{morph}} = 1 / (1 + \exp(-16 \cdot (\text{NC} - 0.60)))$. Weighted fusion: $0.70 \cdot P_{\text{dl}} + 0.30 \cdot P_{\text{morph}}$.	Reconciles AI features with biological truth. Enforces clinical safety override.
Stage 5: Visual Saliency & Export	Grad-CAM backpropagation on layer 12 + ReportLab PDF generation.	Transparent explainability and auditable medical report generation.

4. Technology Stack & Implementation Frameworks

HemaVision was engineered entirely with open, modern, and production-grade technologies, optimizing for zero license fees, rapid local execution, and low inference latency:

Component Layer	Technologies Used	Design Rationale
Deep Learning & Modeling	PyTorch 2.0+, Torchvision, MobileNetV3-Small	Ultra-lightweight footprint (1.52M parameters, 5.9MB weights). Enables sub-50ms inference on standard CPUs.
Computer Vision & Cytometry	OpenCV (cv2), NumPy, PIL / Pillow	Rapid morphological segmentation, contour extraction, adaptive thresholding, and color deconvolution.
Backend & API Microservice	FastAPI (Python 3.12), Uvicorn ASGI, Pydantic	Asynchronous, high-throughput REST API with automated OpenAPI / Swagger documentation.
Explainable AI (XAI)	Custom Grad-CAM Saliency Engine (PyTorch Hooks)	Generates heatmaps directly from gradient activations without black-box third-party dependencies.
Clinical Report Generation	ReportLab PDF Library, Canvas Flowables	Programmatic vector-based clinical PDF reports adhering to standard medical laboratory templates.
Frontend & Interactive Canvas	Semantic HTML5, Vanilla Modern CSS, JavaScript ES6+	Executive clinical light theme, zero framework bloat, 60fps HTML5 Canvas micro-environment simulation.
Conversational Medical Copilot	Google Gemini 2.5 Flash API + Offline Fallback	Dual-mode patient and physician consultation with 35s timeout and local hematology reasoning.

5. Quantitative Evaluation & Clinical Validation Metrics

The model was rigorously validated on unseen peripheral blood smear test sets from the standardized C-NMC hematology benchmark. In clinical oncology screening, **Sensitivity (Recall)** is the primary objective to eliminate False Negatives (i.e., failing to catch a patient with leukemia):

Diagnostic Metric	Measured Score	Clinical Benchmark Interpretation
Overall Accuracy	96.5%	High concordance with senior hematopathologist ground-truth consensus.
Sensitivity (Recall)	97.8%	Critical safety metric: Only 1 out of 50 positive blast cases missed during testing.
Specificity	95.2%	Accurately rules out leukemia in healthy lymphocytes, minimizing unnecessary bone marrow biopsies.
Precision (PPV)	96.1%	When the model flags ALL, the finding is true in >96% of instances.
F1-Score	96.9%	Harmonic mean demonstrating robust performance across imbalanced classes.

ROC-AUC Score	0.988	Near-perfect area under the Receiver Operating Characteristic curve.
Inference Latency	38 ms / cell	Evaluated on Apple Silicon CPU; enables real-time 25 FPS live microscope streaming.

6. Future Scope & Engineering Roadmap

HemaVision has been architected to serve as the baseline for translational medical device certification and clinical deployment:

- **Expansion to 5-Class Hematologic Neoplasm Classifier:** Future releases will train unified vision transformer (ViT) backbones to differentiate ALL, AML (Auer rods), CLL (smudge cells), CML (granulocytic hyperplasia), and Multiple Myeloma in a single forward pass.
- **Whole Slide Imaging (WSI) Gigapixel Tiling:** Implementation of automated gigapixel WSI patch-extraction with Attention-based Multiple Instance Learning (CLAM) to scan entire glass slides automatically.
- **Edge Microfluidic & Smartphone Microscopy Integration:** Porting the quantized MobileNetV3 model onto Raspberry Pi / Jetson Nano devices connected to 3D-printed inverted optical field microscopes for rural screening.
- **FHIR & DICOM Hospital Interoperability:** Direct integration with electronic health records (Epic, Cerner) via standard HL7/FHIR messaging to automatically attach Grad-CAM reports to patient charts.
- **Federated Multi-Hospital Learning:** Deploying privacy-preserving decentralized learning so hospitals can co-train the leukemia model across institutions without sharing raw patient microscopic data.

7. Quickstart Guide: How to Run the Project

The repository includes an automated bootstrap script allowing examiners to launch the entire platform with two simple terminal commands:

```
# 1. Clone repository
git clone https://github.com/Rekha-1kumari/Leukemia-Detection-ML-Pipeline.git
cd Leukemia-Detection-ML-Pipeline

# 2. Create and activate virtual environment
python3 -m venv venv && source venv/bin/activate

# 3. Install production dependencies
pip install -r requirements.txt

# 4. Launch full platform (Model + API + Web UI)
python3 start.py
# Access live application in browser at: http://127.0.0.1:8000
```

CLINICAL DISCLAIMER & CITATION: HemaVision™ is an academic research and educational decision-support pipeline developed by Rekha Kumari. It is engineered to assist healthcare personnel in screening workflows. Final clinical diagnosis and chemotherapy prescriptions must always be established by a board-certified medical oncologist or hematopathologist.